

Stack & Queue Implementation using Structures (Self-Learning)

Q. Why is it advantageous to implement Stacks and Queues using Structures in C? Provide a C program demonstrating the implementation of a Stack and a Queue where their components are encapsulated within a structure.

> **Definition to Remember**

1. Advantages of Using Structures

- **Encapsulation:** Data array and control variables are logically grouped together.

2. Stack Implementation using Structure

```
```c
```

```
// Define the Stack Structure
```

```
// Initialize stack
```

```
// Push operation
```

### 3. Queue Implementation using Structure

```
```c
```

```
// Define the Queue Structure
```

typede

// Initialize queue

void in

// Enqueue operation

void en

> **Must-Write Points (for 10 marks)**

> 1. St

> **Quick Recall**

> `Imp

Pass-k